

1. Solution:

$$2\pi r h = 2\pi r \frac{60}{\pi r^2} = \frac{120}{r}$$

$$M = 2\pi r^2 + \frac{120}{r}$$

$$\frac{dM}{dr} = 4\pi r - \frac{120}{r^2} = 0 \Rightarrow 4\pi r = \frac{120}{r^2}$$

$$\pi r^3 = 30 \quad r = \left(\frac{30}{\pi}\right)^{1/3} = 2.12$$

$$\text{Glob Minimum} = 2.12$$

2. Solution

$$2x + 2y = 72, y = 36 - x$$

$$0 \leq x \leq 36$$

$$A = xy = x(36 - x) = 36x - x^2$$

$$\frac{dA}{dx} = 36 - 2x = 0, x = \frac{36}{2} = 18$$

$$x = 18, \text{ ENDPOINTS: } x = 0, 36$$

$$x = 0, A = 0 \text{ cm}^2$$

$$x = 18, A = 18(36 - 18) = 324 \text{ cm}^2$$

$$x = 36, A = 36(36 - 36) = 0$$

Max occurs at 18 cm^2

3. Solution:

$$X+Y=85 \quad Y=85-X$$

$$0 \leq X \leq 85$$

$$P=XY=X(85-X)=85X-X^2$$

$$\frac{dP}{dX}=85-2X=0$$

$$X=\frac{85}{2}=42.5$$

$$X=42.5$$

$$X=0, 85$$

$$X=0 \quad P=0$$

$$X=42.5, P=42.5(85-42.5)=1806.25$$

$$X=85, P=85(85-85)=0$$

$$\text{Max}=1806.25$$

$$XY=492$$

$$2=X+1$$

$$2=X+492/X$$

$$0=1-492/X^2$$

$$-1=-492/X^2$$

$$X^2=492$$

$$22.18$$

$$\text{Max}=1806.25$$

4. Solution:

$$Y = x^2 e^{-6x}, \text{ for } x \geq 0$$

$$\text{Slope} = \frac{Y}{x} = \frac{x^2 e^{-6x}}{x} = x e^{-6x}$$

$$\frac{d}{dx} = 1 \cdot e^{-6x} - 6x e^{-6x} = 0$$

$$e^{-6x} (1 - 6x) = 0$$

$$x = \frac{1}{6}$$

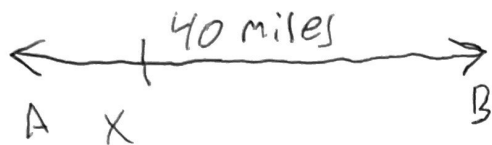
$$\text{Slope} = \frac{1}{6} e^{-6(1/6)} = \frac{1}{6} e^{-1} = \frac{1}{6e}$$

$$\text{Max Slope} = 1/6e \text{ at } 1/6$$

5. Solution:

~~D~~

$$D = \frac{k_1}{x^2} + \frac{k_2}{(40-x)^2}$$



$$S = \frac{k_1}{x^2} + \frac{k_2}{(40-x)^2} = k^2 \left(\frac{14}{x^2} + \frac{1}{(40-x)^2} \right)$$

$$f(x) = 14x^{-2} + (40-x)^{-2}$$

~~$$f'(x) = -28x^{-3} - 2(40-x)^{-3}(-1) = -\frac{28}{x^3} + \frac{2}{(40-x)^3}$$~~

$$-\frac{28}{x^3} + \frac{2}{(40-x)^3} = \frac{-28(40-x)^3 + 2x^3}{x^3(40-x)^3}$$

$$-28(40-x)^3 + 2x^3 = 0$$

$$2x = 28(40-x)^3$$

$$x^3 = 14(40-x)^3$$

$$\frac{40-x}{x} = (1/14)^{1/3} = 0.411$$

$$40-x = 0.411x, x = 40 / 1.411 = 28.36$$

$$0 < x < 40$$

$$f''(x) = 84x^{-4} + 6(40-x)^{-4} = \frac{84}{x^4} + \frac{6}{(40-x)^4} > 0$$

Ans: Minimum is at 28.36 miles

6. Solution:

$$2x + y = 600 \quad y = 600 - 2x$$

$$A(x) = x(600 - 2x) = 600x - 2x^2$$

$$A'(x) = 600 - 4x, \quad 600 - 4x = 0 \quad x = 150$$

$$y = 411 - x^2$$

$$A(x) = 2x(411 - x^2) = 822x - 2x^3$$

$$A'(x) = 822 - 6x^2 \rightarrow 822 - 6x^2 = 0 \rightarrow x^2 = 36$$

$$36 - x^2$$

$$A(x) = 2x(36 - x^2) = 72x - 2x^3$$

$$A'(x) = 72 - 6x^2 \rightarrow 72 - 6x^2 = 0 \Rightarrow x^2 = 12 \rightarrow x = 3.46$$

$$A''(x) = -12x$$

$$A''(3.46)$$

$$72 - 6x^2 = 0$$

$$x = 12 \quad x = 3.46$$

$$A = 72 \cdot 3.46 - 2 \cdot 36$$

$$177.12 \text{ Sa units}$$

7. Solution:

$$x + 2y = 72$$

$$f(x) = (x+1)(y+2)$$

$$x = 72 - 2y \rightarrow f(y) = (72 - 2y + 1)(y + 2) \Rightarrow$$

$$(73 - 2y)(y + 2) = 146 + 69y - 2y^2$$

$$69 - 4y \rightarrow 69 - 4y = 0 \Rightarrow y = \frac{69}{4}$$

$$72 - 2\left(\frac{69}{4}\right) = 37.5$$

$$x = 37.5 \quad y = 17.25$$